Juan A. Fraire

Researcher in Space Networking, AI & Orbital Computing

IRTF SPACE RG Chair • STINT Co-Founder & Chair • Inria, CONICET & Saarland University

juan.fraire@inria.fr • juanfraire.space • ORCID • Google Scholar • DBLP • Inria Page

Argentinian & Italian (EU) citizen

Short Bio: I am a researcher at Inria (France) and a guest researcher and professor at CONICET-UNC (Argentina) and Saarland University (Germany). My work bridges telecommunications engineering and computer science to advance space networking, orbital edge/cloud computing, and onboard AI, treating orbital dynamics, intermittent connectivity, energy, and compute as first-class design constraints. I focus on near-Earth and deep-space network analysis, design, and operation, which have resulted in more than 180 published papers in international journals and leading conferences. I am the chair of the Systems and Protocol Aspects for Circumstellar Environments RG (SPACE) at the IRTF and the co-founder and chair of the Space-Terrestrial Internetworking Workshop (STINT). I actively lead and participate in diverse joint projects with space agencies (e.g., NASA, ESA, CONAE) and companies in the space sector (e.g., D3TN, Skyloom).

At a Glance 180+ papers, books & chapters | 3,000+ citations, h-index 29, i10-index 76 (Google Scholar) | €7M+ in funded projects | 16 doctoral & 25+ master's students | 13 STINT editions chaired

Appointments

Academic

- **Inria Starting Faculty Position (ISFP)**, Institut National de Recherche en Informatique et en Automatique (Inria), Lyon, France — January 2021 – Present
  - Research Topic: *Protocols and algorithms for space-terrestrial integrated IoT networks.*
  - Main project: ANR PRCE Space-Terrestrial Integrated Internet of Things (STEREO).
  - Habilitation to Direct Research (HDR): Defended on May 2025.
- **Assistant Researcher (on continuous detachment)**, National Council of Scientific and Technical Research (CONICET), Córdoba, Argentina — July 2017 – Present
  - Research Topic: *Protocols and algorithms for space-terrestrial integrated networks.*
  - Mobility:
    * TIMA Laboratory (France), visiting researcher, 6 months.
    * Saarland University (Germany), visiting researcher, 2 months.
    * JPL, NASA (USA), Fulbright visiting scholar program, 3 months.
    * Nanjing University (China), visiting researcher, 3 months.
    * MISSION (EU RISE project), national coordinator, 5 years.
  - Transfers:
    * Skyloom (California, USA): Geostationary full-optical relay.
    * D3TN (Dresden, Germany): ColdSun (ESA), Orbitare, and DARKSOL (EU/Saxony space cloud) projects.
    * Cloud and Heat (Dresden, Germany): Aocloud project.
- **Guest Professor**, Dependable Systems and Software, Saarland University, Saarbrücken, Germany — Since September 2018
  - Formal member of Prof. Holger Hermanns' group, bridging space networking and formal dependability.
- **Postdoctoral Researcher**, Politecnico di Torino (DET Department), Torino, Italy — 2020 – 2021
  - Research on satellite network architectures and routing, held while on detachment from CONICET (before joining Inria).
- **Postdoctoral Researcher**, Université Grenoble Alpes (UGA), TIMA Laboratory, Grenoble, France

June 2016 – September 2017
  - Research Topic: *Delay and Fault Tolerant Mobile Communications Architectures for Future Satellite Networks*. Supervised by Dr. Raoul Velazco. Funded by project CAPACITES.

Industry & Entrepreneurial

- **Co-Founder & Chief Executive Officer**, D3TN U.S., Florida, USA — Since 2022
  - Specialized space networking and protocol design consultancy services for the US market.
- **Co-Founder & Chief Scientific Officer**, Pontos, Bordeaux, France — Since 2024
  - Research and development of advanced AI and satellite-terrestrial networking solutions for maritime equipment.
- **Mobile Networks Planning Engineer**, Nokia, Córdoba, Argentina — August 2011 – April 2012
  - Planning and configuration of 2G and 3G networks for commercial deployments in Latin America; management of regional outsourcing.
- **LTE Field and Trial Engineer**, Motorola, Córdoba, Argentina — June 2010 – July 2011
  - New Product Introduction (NPI) Engineer for 3GPP LTE; LTE network feature planning and worldwide Motorola staff training.
- **WiMAX Field and Trial Engineer**, Motorola, Córdoba, Argentina — June 2008 – May 2010
  - New Product Introduction (NPI) Engineer for IEEE 802.16 (WiMAX); feature validation with external customers and WiMAX Forum certification.

Education

- **Habilitation à Diriger des Recherches (HDR)**, INSA Lyon / Université Claude Bernard Lyon 1, Lyon, France — May 2025
  - Title: *Space Networking: Models and Methods*
  - Jury: Prof. Petar Popovski (Aalborg University), Prof. Elif Uysal (METU), Prof. Gunes Karabulut Kurt (Polytechnique Montréal), Prof. Sylvia Ratnasamy (UC Berkeley), Prof. Hervé Rivano (INSA Lyon), Prof. Scott Shenker (UC Berkeley), Prof. Isabelle Guérin Lassous (Université Claude Bernard Lyon 1).
- **Abilitazione Scientifica Nazionale (ASN)**, Ministero dell'Università e della Ricerca, Italy — June 2026
  - National qualification for the functions of Associate Professor (*II fascia*), Settore Concorsuale 09/F2 — Telecommunications (call ASN 2023/2025, D.D. n. 1796/2023). Valid until June 2038.
  - Sector 09/F2 is bibliometric: qualification requires exceeding national median thresholds for normalized publications, citations and h-index, verified by the ministerial commission on Scopus and Web of Science.
- **Ph.D. in Applied Sciences**, Universidad Nacional de Córdoba (UNC), Argentina — October 2015
  - Thesis: *Contact Plan Design for Disruption Tolerant Satellite Networks*
  - Context: Network Architect of CONAE's (Argentinian Space Agency) SARE satellite constellation, coordinating a team of researchers at the UNC.
  - Advisors: Dr. Pablo A. Ferreyra and Dr. Jorge M. Finochietto.
- **Diploma in Data Science and Machine Learning**, Universidad Nacional de Córdoba (UNC), Argentina — December 2022
- **Telecommunications Engineering**, Instituto Universitario Aeronáutico (IUA), Argentina — GPA 8.69/10 — December 2008
- **Cisco Certified Network Associate (CCNA)**, Cisco Networking Academy (Proydesa), Argentina — GPA 9.00/10 — January 2008

Honors & Awards

- **Best Engineering Ph.D. Graduate Award**, Academia Nacional de Ingeniería, Argentina (2015).
- **Fulbright Visiting Scholar**, JPL/NASA, USA — competitive research fellowship.

- **Best Engineer Graduate Award**, Academia Nacional de Ingeniería, Argentina (2008).
- **Best Career Average: Golden Medal**, Instituto Universitario Aeronáutico (2008).
- **Best Career Average Scholarship Award**, Motorola (2008).
- **Córdoba Best Student Award**, Ministerio de Educación, Argentina.
- **CCNA Best Average: Golden Medal**, Cisco Networking Academy (2008).

Funding & Grants

Participated in more than 15 competitively funded projects totaling over €7M as coordinator, principal investigator, or partner.

- **Active Projects:**
  - **STEREO** (Space-Terrestrial Integrated IoT): ANR (France), PRCE, €1M (2023–2027). *Coordinator*. Partners: Inria, IRIT, LAAS, LIG, Kineis.
  - **MISSION** (Models in Space Systems): EU RISE, €1.5M (2021–2026). *Responsible Researcher*. Partners: U. Twente, RWTH Aachen, Saarland U., UNC, UNRC, D3TN, Ascentio, INVAP.
  - **DONUTS** (Design of Multiscale Space Networks): France 2030, €2.4M (2025–2029). *Contributor*. Partners: U. Toulouse, U. Strasbourg, ENS Lyon, INSA Lyon, Grenoble INP, LAAS, CEA.
  - **VISTA** (Viable and Scalable Interplanetary and Terrestrial Architectures): ANR (France), JCJC (2026–2030). *Collaborator*. Coordinated by Dr. Olivier De Jonckère (LIRMM).
  - **D3-CONNECT** (Direct-to-Device Connectivity): Inria Associate Team with UC Berkeley, €32k (2025–2027). *Principal Investigator* (with Prof. Sylvia Ratnasamy, UC Berkeley NetSys Lab).
  - **DORSAL-IoT** (Direct-to-Satellite LoRaWAN): STIC-AmSud, €33k (2025–2027). *Coordinator Group Member*. Partners: Inria, U. de Chile, UDP, UTEM, UFSC (Brazil).
  - **MOBIDEC** (PEPR, targeted project PC3): France 2030, €89k allocated to Inria Agora (to 2028). *Contributor*.
  - **ConOpsCon** (Constellations Operation with Continuous Communication): ESA EXPRO+, €300k (contract 4000152960, 2026–). *Coordinator* (Saarland University as prime contractor). Partner: VisionSpace Technologies.
- **Doctoral Scholarships Secured:**
  - **ED Lyon Ph.D. Scholarship:** €36k/year (2022–2025). Ph.D.: Diego Maldonado.
  - **CONICET Ph.D. Scholarship:** ARS-funded (2022–2025). Ph.D.: Santiago Henn.
  - **UNIGE Ph.D. Scholarship:** Italian funds (2023–2027). Ph.D.: Camilo Rojas.
  - **CIFRE Ph.D. with Kinéis:** industrial doctoral contract (2025–2028), with IRIT/Toulouse INP. Ph.D.: Benoit Coeugnet.
  - **ED InfoMaths Ph.D. Scholarship + ESA OSIP co-funding:** INSA doctoral contract (2026–2029) co-funded by ESA (Cooperative Agreement 4000153483, *A Digital Twin for Optimization and Operations in the Solar System Internet*). Ph.D.: Alice Le Bihan.
- **Concluded Projects:**
  - **AI4LFNM** (AI for Fleet Network Management): ESA, €500k (2023–2025). *Lead Researcher*. Partners: GMV (prime), Saarland U., Thales Alenia Space, Eutelsat, Planet Labs.
  - **ML4OISL** (Machine Learning for Optical ISL): ESA, €500k (2024–2025). *Contributor*. Partners: GMV (prime), Saarland U., Eutelsat.
  - **HANDING-OVER** (Handover, Routing & Resource Management for VLEO): ESA, €500k (2023–2025). *Contributor*. Partners: Thales Alenia Space (prime), Politecnico di Torino.
  - **i2Cat Contract:** €50k (2023). Optical ISLs and network management.
  - **NEON** (Network of Competences on IoT): Erasmus+, €1M (2021–2024). *Collaborator*.
  - **STARS** (Satellite Networks Protocols): STIC-AmSud, €30k (2021–2023). *Responsible Researcher*.
  - **ContactAR** (Contact Tracing): ANPCyT, €150k (2020–2021). *Responsible Researcher*.
  - **MAPS-Net** (Segmented Satellite Networks): ANPCyT, €60k (2019–2021). *Director*.
  - **LEOPowVer**: ERC Proof of Concept, €150k (2021–2022). *PI Adjunct* (with Prof. H. Hermanns);

in-orbit validation on GomSpace's GomX-4.

Selected Publications

[1] Robin Ohs, Gregory F. Stock, Andreas Schmidt, **Juan A. Fraire**, Jörg Ott, and Holger Hermanns. "Dark Clouds Rising in Low-Earth Orbit: On Environmental Limits to Massive Orbital AI". In: *ACM Workshop on LEO Networking and Communication (LEO-NET), co-located with ACM SIGCOMM*. 2026, pp. 1960–1965. DOI: 10.1145/3789240.3827598.

[2] Alexander Ylnner Choquenaira Florez, **Juan A. Fraire**, Hannah B. Pasandi, and Hervé Rivano. "On the role of machine learning in satellite internet of things: A survey of techniques, challenges, and future directions". In: *Elsevier Computer Networks* 259 (2025), p. 111063. DOI: 10.1016/j.comnet.2025.111063.

[3] Gabriel Maiolini Capez, Mauricio A. Cáceres, Roberto Armellin, Christopher P. Bridges, **Juan A. Fraire**, Stefan Frey, and Roberto Garello. "On the Use of Mega Constellation Services in Space: Integrating LEO Platforms Into 6G Non-Terrestrial Networks". In: *IEEE Journal on Selected Areas in Communications* 42.12 (2024), pp. 3490–3504. DOI: 10.1109/jsac.2024.3459078.

[4] Gabriel Maiolini Capez, Santiago Henn, **Juan A. Fraire**, and Roberto Garello. "Sparse Satellite Constellation Design for Global and Regional Direct-to-Satellite IoT Services". In: *IEEE Transactions on Aerospace and Electronic Systems* 58.5 (2022), pp. 3786–3801. DOI: 10.1109/taes.2022.3185970.

[5] **Juan A. Fraire**, Oana Iova, and Fabrice Valois. "Space-Terrestrial Integrated Internet of Things: Challenges and Opportunities". In: *IEEE Communications Magazine* 60.12 (2022), pp. 64–70. DOI: 10.1109/mcom.008.2200215.

[6] Gregory Stock, **Juan A. Fraire**, and Holger Hermanns. "Distributed On-Demand Routing for LEO Mega-Constellations: A Starlink Case Study". In: *Advanced Satellite Multimedia Systems Conference and Signal Processing for Space Communications Workshop (ASMS/SPSC)*. IEEE, 2022, pp. 1–8. DOI: 10.1109/asms/spsc55670.2022.9914716.

[7] **Juan A. Fraire**, Olivier De Jonckère, and Scott C. Burleigh. "Routing in the Space Internet: A contact graph routing tutorial". In: *Elsevier Journal of Network and Computer Applications* 174 (2021), p. 102884. DOI: 10.1016/j.jnca.2020.102884.

[8] **Juan A. Fraire**, Sandra Céspedes, and Nicola Accettura. "Direct-To-Satellite IoT - A Survey of the State of the Art and Future Research Perspectives - Backhauling the IoT Through LEO Satellites". In: *International Conference on Ad-Hoc Networks and Wireless (ADHOC-NOW)*. Ed. by Maria Rita Palattella, Stefano Scanzio, and Sinem Coleri Ergen. Vol. 11803. Lecture Notes in Computer Science. Springer, 2019, pp. 241–258. DOI: 10.1007/978-3-030-31831-4_17.

[9] **Juan A. Fraire**, Jorge M Finochietto, and Scott C Burleigh. *Delay Tolerant Satellite Networks*. Artech House, 2017. ISBN: 9781630815172.

[10] **Juan A. Fraire**, Pablo G. Madoery, Scott C. Burleigh, Marius Feldmann, Jorge M. Finochietto, Amir Charif, Nacer-Eddine Zergainoh, and Raoul Velazco. "Assessing Contact Graph Routing Performance and Reliability in Distributed Satellite Constellations". In: *Journal of Computer Networks and Communications* 2017 (2017), 2830542:1–2830542:18. DOI: 10.1155/2017/2830542.

[11] **Juan A. Fraire** and Jorge M. Finochietto. "Design challenges in contact plans for disruption-tolerant satellite networks". In: *IEEE Communications Magazine* 53.5 (2015), pp. 163–169. DOI: 10.1109/mcom.2015.7105656.

[12] **Juan A. Fraire** and Jorge M. Finochietto. "Routing-aware fair contact plan design for predictable delay tolerant networks". In: *Elsevier Ad Hoc Networks* 25 (2015), pp. 303–313. DOI: 10.1016/j.adhoc.2014.07.006.

Research Software & Deployment

- **In orbit**: my contact-graph routing modules run in **NASA's ION and HDTN** DTN stacks (ISS routing-experiment payload and Deep Space Network stations); battery-aware routing and autonomous inter-satellite-link scheduling were demonstrated on **GomSpace's GomX-4** LEO satellites; BPv7 / Ring Road Networks were validated on **ESA's OPS-SAT** mission.
- **μD3TN**: participated in the development of the lightweight, software-defined DTN protocol stack for constrained spacecraft (space and underwater deployments).
- **Open-source & research tools**: DtnSim, pyCGR (CGR library adopted in NASA's HDTN), FLoRaSat / FLoRaSat 2, MeteorNet (Space Cloud emulation), A-SABR (schedule-aware bundle routing), IPN-V (interplanetary network visualizer), and Contact Plan Designer (STK/Ansys plug-in).
- **Archived software (Zenodo)**: ESpaS, Estimator for Space Sustainability (2025, DOI: 10.5281/zenodo.15745708); PhantomLink, emulation of virtual end-to-end links on ground and in orbit (2025, DOI: 10.5281/zenodo.15680651); and Uncertainty-Aware Routing in Delay-Tolerant Networks, artifact and reproducibility package (2026, DOI: 10.5281/zenodo.18507793).

Publications

Books and Chapters

[1] **Juan A. Fraire** and Jorge M Finochietto. "DTN of Things". In: *Delay and Disruption Tolerant Networks (CRC Press)*. CRC Press, 2018, pp. 275–308.

[2] **Juan A. Fraire**, Jorge M Finochietto, and Scott C Burleigh. *Delay Tolerant Satellite Networks*. Artech House, 2017. ISBN: 9781630815172.

[3] **Juan A. Fraire**, Pablo G. Madoery, and Jorge M. Finochietto. "Contact Plan Design for Predictable Disruption-Tolerant Space Sensor Networks". In: *Wireless Sensor Systems for Extreme Environments: Space, Underwater, Underground and Industrial*. Ed. by Habib F. Rashvand and Ali Abedi. Wiley, 2017, pp. 123–150. DOI: 10.1002/9781119126492.ch7.

Journal Articles

[1] Serdar Akar, Alessandro Compagnoni, **Juan A. Fraire**, and Roberto Garello. "Direct-to-satellite internet of things (DtS-IoT): a tutorial review on architectures, protocols, and future directions". In: *Frontiers in Communications and Networks* 7 (2026). DOI: 10.3389/frcmn.2026.1750955.

[2] Elena Fernández-Niño, **Juan A. Fraire**, Adriano Camps, and Joan A. Ruiz-de-Azua. "RF-Assisted Compensation for Rapid Optical Acquisition in Hybrid Inter-Satellite Links". In: *IEEE Open Journal of the Communications Society* 7 (2026), pp. 6625–6639. DOI: 10.1109/ojcoms.2026.3688693.

[3] **Juan A. Fraire**. "Powering the Cloud Above the Clouds: Are Orbital Data Centers the First Real Customer for Space-Based Solar Power?" In: *IEEE Aerospace and Electronic Systems Magazine* (2026). Early access, pp. 1–10. DOI: 10.1109/maes.2026.3742742.

[4] **Juan A. Fraire**, Benoit Coeugnet, Oana Iova, André-Luc Beylot, and Vincent Deslandes. "Blind vs. Satellite-Detect Uplink in LEO Direct-to-Satellite IoT: An Analytical Threshold". In: *IEEE Transactions on Vehicular Technology* (2026). Early access. DOI: 10.1109/tvt.2026.3733173.

[5] **Juan A. Fraire**, Oana Iova, Carlos Fernandez Hernandez, Fabrice Valois, Hervé Rivano, Cesar Azurdia-Meza, Marcos Diaz, Miguel Gutiérrez Gaitán, Diego Dujovne, Samuel Montejo Sánchez, and Richard Demo Souza. "DORSAL: Downlink Optimization for Robust Direct-to-Satellite LoRaWAN". In: *IEEE Internet of Things Journal* 13.15 (2026), pp. 35318–35331. DOI: 10.1109/jiot.2026.3696890.

[6] Wenxiao Ge, **Juan A. Fraire**, Xinxin Shen, and Kanglian Zhao. "Space Cloud Networks: Technologies, Architectures, and Key Enablers". In: *IEEE Communications Magazine* 64.6 (2026), pp. 30–36. DOI: 10.1109/mcom.001.2500427.

[7] Wenxiao Ge, **Juan A. Fraire**, and Kanglian Zhao. “Optimal Design of Space-Based Data Center Constellations for In-Orbit Computing Services”. In: *IEEE Communications Letters* 30 (2026), pp. 1880–1884. DOI: 10.1109/lcomm.2026.3688042.

[8] Jason Gerard, **Juan A. Fraire**, and Sandra Céspedes. “Contact Plan Design for Optical Interplanetary Communications”. In: *Elsevier Ad Hoc Networks* 194 (2026), p. 104393. DOI: 10.1016/j.adhoc.2026.104393.

[9] Camilo Rojas Milla, Fabio Patrone, **Juan A. Fraire**, and Mario Marchese. “From emerging LEO satellite constellations to the space cloud: Emulation platforms and orchestration methods”. In: *Elsevier Computer Networks* 276 (2026), p. 111970. DOI: 10.1016/j.comnet.2025.111970.

[10] Arnau Singla, Anna Calveras, **Juan A. Fraire**, and Joan A. Ruiz-de-Azua. “End-to-End Testbed Validation of Periodic and Probabilistic-Aware Constellation Scheduling for Non-Terrestrial Networks”. In: *Elsevier Acta Astronautica* 250 (2026), pp. 69–83. DOI: 10.1016/j.actaastro.2026.08.074.

[11] Huan Yan, **Juan A. Fraire**, Ziqi Yang, Kanglian Zhao, Wenfeng Li, Xiyun Hou, Haohan Li, Yuxuan Miao, Jinjun Zheng, Chengbin Kang, Huichao Zhou, Xinuo Chang, Lu Wang, and Linshan Xue. “Contact Plan Design for Cross-Linked GNSSs: An ILP Approach for Extended Applications”. In: *IEEE Transactions on Aerospace and Electronic Systems* 62 (2026), pp. 6751–6770. DOI: 10.1109/taes.2026.3671753.

[12] Huan Yan, **Juan A. Fraire**, Ziqi Yang, Kanglian Zhao, Wenfeng Li, Xiyun Hou, Haohan Li, Yuxuan Miao, Jinjun Zheng, Chengbin Kang, Huichao Zhou, Xinuo Chang, Lu Wang, and Linshan Xue. “Joint Contact Planning for Navigation and Communication in GNSS-Libration Point Systems”. In: *IEEE Transactions on Vehicular Technology* 75.5 (2026), pp. 8565–8580. DOI: 10.1109/tvt.2025.3632905.

[13] Huan Yan, **Juan A. Fraire**, Ziqi Yang, Kanglian Zhao, Wenfeng Li, Jinjun Zheng, Chengbin Kang, Huichao Zhou, Xinuo Chang, Lu Wang, Linshan Xue, Tatiana Podladchikova, Maria Kudryashova, and Zaina Abushaar. “Optimized Contact Plan Design for Reflector and Phased Array Terminals in Cislunar Space Networks”. In: *Science China Information Sciences* 69.11 (2026), p. 219301. DOI: 10.1007/s11432-025-4892-7.

[14] Lei Yang, Kanglian Zhao, **Juan A. Fraire**, Ruhai Wang, Scott C. Burleigh, Wenfeng Li, and Hong Yang. “Semicentralized Online Resource Allocation for Ensuring QoS in Deep Space DTN”. In: *Wiley International Journal of Satellite Communications and Networking* 44.2 (2026), pp. 184–198. DOI: 10.1002/sat.70012.

[15] Alexander Y. Choquenaira-Florez, Benoit Coeugnet, Robin Ohs, **Juan A. Fraire**, and Hervé Rivano. “FLoRaSat 2: Simulating Cross-Linked Direct-to-Satellite IoT LEO Constellations with Joint Access and Routing Evaluation”. In: *Wiley International Journal of Satellite Communications and Networking* (2025). In press. URL: https://hal.science/hal-05424077.

[16] Pedro R. D’Argenio, **Juan A. Fraire**, Arnd Hartmanns, and Fernando D. Raverta. “Comparing Statistical, Analytical, and Learning-Based Routing Approaches for Delay-Tolerant Networks”. In: *ACM Transactions on Modeling and Computer Simulation* 35.2 (2025), 10:1–10:26. DOI: 10.1145/3665927.

[17] Alexander Ylnner Choquenaira Florez, **Juan A. Fraire**, Hannah B. Pasandi, and Hervé Rivano. “On the role of machine learning in satellite internet of things: A survey of techniques, challenges, and future directions”. In: *Elsevier Computer Networks* 259 (2025), p. 111063. DOI: 10.1016/j.comnet.2025.111063.

[18] Diego Maldonado, Leonardo S. Cardoso, **Juan A. Fraire**, Alexandre Guitton, Oana Iova, Megumi Kaneko, and Hervé Rivano. “Enhanced LR-FHSS receiver for headerless frame recovery in space-terrestrial integrated IoT networks”. In: *Elsevier Computer Networks* 257 (2025), p. 111018. DOI: 10.1016/j.comnet.2024.111018.

[19] Diego Maldonado, Megumi Kaneko, **Juan A. Fraire**, Alexandre Guitton, Oana Iova, and Hervé Rivano. “Enhancing LR-FHSS Scalability Through Advanced Sequence Design and Demodu-

lator Allocation". In: *IEEE Transactions on Green Communications and Networking* 9.4 (2025), pp. 1716–1729. DOI: 10.1109/tgcn.2025.3570185.

[20] Camilo Rojas Milla, Nour Badini, Fabio Patrone, **Juan A. Fraire**, and Mario Marchese. "Quantifying Edge Server Placement in LEO Constellations: A Comparative Emulation of Space and Terrestrial Cloud". In: *Wiley International Journal of Satellite Communications and Networking* (2025). In press. URL: https://hal.science/hal-05424023v1.

[21] Arnau Singla, Franco Criscola, David Canales, **Juan A. Fraire**, Anna Calveras, and Joan A Ruiz-de-Azua. "Toward Integrated Satellite Operations and Network Management: A Review and Novel Framework". In: *MDPI Technologies* 13.8 (2025), p. 312. DOI: 10.3390/technologies13080312.

[22] Arnau Singla, Isabel Gallardo-Duval, Anna Calveras, **Juan A. Fraire**, and Joan Adrià Ruiz-de-Azua. "Probabilistic-Aware Satellite Constellation Scheduling for Integrated TN-NTN Operations". In: *IEEE Open Journal of the Communications Society* 6 (2025), pp. 3950–3963. DOI: 10.1109/ojcoms.2025.3564411.

[23] Gabriel Maiolini Capez, Mauricio A. Cáceres, Roberto Armellin, Christopher P. Bridges, **Juan A. Fraire**, Stefan Frey, and Roberto Garello. "On the Use of Mega Constellation Services in Space: Integrating LEO Platforms Into 6G Non-Terrestrial Networks". In: *IEEE Journal on Selected Areas in Communications* 42.12 (2024), pp. 3490–3504. DOI: 10.1109/jsac.2024.3459078.

[24] Elena Fernández-Niño, **Juan A. Fraire**, Adriano Camps, and Joan Adrià Ruiz-de-Azua. "RF-Assisted Uncertainty Cone Reduction in Free-Space Optical Inter-Satellite Links". In: *IEEE Open Journal of the Communications Society* 5 (2024), pp. 7598–7612. DOI: 10.1109/ojcoms.2024.3507555.

[25] **Juan A. Fraire**, Santiago Henn, Gregory Stock, Robin Ohs, Holger Hermanns, Felix Walter, Lynn Van Broock, Gabriel Ruffini, Federico Machado, Pablo Serratti, and Jose Relloso. "Quantitative analysis of segmented satellite network architectures: A maritime surveillance case study". In: *Elsevier Computer Networks* 255 (2024), p. 110874. DOI: 10.1016/j.comnet.2024.110874.

[26] **Juan A. Fraire**, Joan Adrià Ruiz-de-Azua, and Elena Fernández-Niño. "Toward Optimal Placement of Free-Space Optical Terminal on the Spacecraft Body". In: *IEEE Transactions on Aerospace and Electronic Systems* 60.6 (2024), pp. 8914–8928. DOI: 10.1109/taes.2024.3435176.

[27] Oscar Ledesma, Paula Lamo, and **Juan A. Fraire**. "Trends in LPWAN Technologies for LEO Satellite Constellations in the NewSpace Context". In: *MDPI Electronics* 13.3 (2024), p. 579. DOI: 10.3390/electronics13030579.

[28] Oscar Ledesma, Paula Lamo, **Juan A. Fraire**, María Ruiz, and Miguel A Sánchez. "Architectural framework and feasibility of internet of things-driven mars exploration via satellite constellations". In: *MDPI Electronics* 13.7 (2024), p. 1289. DOI: 10.3390/electronics13071289.

[29] Raydel Ortigueira, Samuel Montejo-Sánchez, Santiago Henn, **Juan A. Fraire**, and Sandra Céspedes. "Satellite Visibility Prediction for Constrained Devices in Direct-to-Satellite IoT Systems". In: *IEEE Sensors Journal* 24.16 (2024), pp. 26630–26644. DOI: 10.1109/jsen.2024.3418728.

[30] Hongcheng Yan, Liang Qiao, Wei Wu, **Juan A. Fraire**, Dong Zhou, Luming Li, and Yong Xu. "Routing in future space-terrestrial integrated networks with SATNET-OSPF". In: *Wiley International Journal of Satellite Communications and Networking* 42.1 (2024), pp. 1–25. DOI: 10.1002/sat.1495.

[31] Lei Yang, **Juan A. Fraire**, Kanglian Zhao, Ruhai Wang, Wenfeng Li, and Hong Yang. "Optimizing deep-space DTN congestion control via deep reinforcement learning". In: *Elsevier Computer Networks* 255 (2024), p. 110865. DOI: 10.1016/j.comnet.2024.110865.

[32] Renato Cherini, Ramiro Detke, **Juan A. Fraire**, Pablo G. Madoery, and Jorge M. Finochietto. "Toward Deep Digital Contact Tracing: Opportunities and Challenges". In: *IEEE Pervasive Computing* 22.4 (2023), pp. 15–25. DOI: 10.1109/mprv.2023.3320987.

[33] Santiago Henn, **Juan A. Fraire**, and Holger Hermanns. “Polygon-Based Algorithms for N-Satellite Constellations Coverage Computing”. In: *IEEE Transactions on Aerospace and Electronic Systems* 59.5 (2023), pp. 7166–7182. DOI: 10.1109/taes.2023.3289479.

[34] Pablo Ilabaca Parra, Samuel Montejo Sanchez, **Juan A. Fraire**, Richard Demo Souza, and Sandra Céspedes. “Network Size Estimation for Direct-to-Satellite IoT”. In: *IEEE Internet of Things Journal* 10.7 (2023), pp. 6111–6125. DOI: 10.1109/jiot.2022.3224678.

[35] Guido Álvarez, **Juan A. Fraire**, Khaled Abdelfadeel Hassan, Sandra Céspedes, and Dirk Pesch. “Uplink Transmission Policies for LoRa-Based Direct-to-Satellite IoT”. In: *IEEE Access* 10 (2022), pp. 72687–72701. DOI: 10.1109/access.2022.3189647.

[36] Gabriel Maiolini Capez, Santiago Henn, **Juan A. Fraire**, and Roberto Garello. “Sparse Satellite Constellation Design for Global and Regional Direct-to-Satellite IoT Services”. In: *IEEE Transactions on Aerospace and Electronic Systems* 58.5 (2022), pp. 3786–3801. DOI: 10.1109/taes.2022.3185970.

[37] Marius Feldmann, **Juan A. Fraire**, Felix Walter, and Scott C. Burleigh. “Ring Road Networks: Access for Anyone”. In: *IEEE Communications Magazine* 60.4 (2022), pp. 38–44. DOI: 10.1109/mcom.001.2100835.

[38] **Juan A. Fraire**, Oana Iova, and Fabrice Valois. “Space-Terrestrial Integrated Internet of Things: Challenges and Opportunities”. In: *IEEE Communications Magazine* 60.12 (2022), pp. 64–70. DOI: 10.1109/mcom.008.2200215.

[39] Andrea Nardin, **Juan A. Fraire**, and Fabio Dovis. “Contact Plan Design for GNSS Constellations: A Case Study With Optical Intersatellite Links”. In: *IEEE Transactions on Aerospace and Electronic Systems* 58.3 (2022), pp. 1981–1995. DOI: 10.1109/taes.2021.3135025.

[40] Jasmine Rimani, Luigi Mascolo, and **Juan A. Fraire**. “A Parametric Data Handling Evaluation Framework for Autonomous Lunar Networks”. In: *Springer CEAS Space Journal* 14.2 (2022), pp. 365–376. DOI: 10.1007/s12567-021-00390-4.

[41] Pablo A. Ferreyra, Rubén Danilo Capkob, Alberto Fabián Gómez, **Juan A. Fraire**, and Carlos José Barrientos. “Embedded wireless delay tolerant networks on chips for segmented architectures”. In: *International Journal of Embedded Systems* 14.6 (2021), pp. 578–591. DOI: 10.1504/ijes.2021.121087.

[42] **Juan A. Fraire** and Juan Eduardo Durán. “Revising Computer Science Networking Hands-On Courses in the Context of the Future Internet”. In: *IEEE Transactions on Education* 64.2 (2021), pp. 133–138. DOI: 10.1109/te.2020.3015673.

[43] **Juan A. Fraire** and Elías L. Gasparini. “Centralized and Decentralized Routing Solutions for Present and Future Space Information Networks”. In: *IEEE Network* 35.4 (2021), pp. 110–117. DOI: 10.1109/mnet.011.2100102.

[44] **Juan A. Fraire**, Carsten Gerstacker, Holger Hermanns, Gilles Nies, Morten Bisgaard, and Kristian Bay. “On the scalability of battery-aware contact plan design for LEO satellite constellations”. In: *Wiley International Journal of Satellite Communications and Networking* 39.2 (2021), pp. 193–204. DOI: 10.1002/sat.1374.

[45] **Juan A. Fraire**, Olivier De Jonckère, and Scott C. Burleigh. “Routing in the Space Internet: A contact graph routing tutorial”. In: *Elsevier Journal of Network and Computer Applications* 174 (2021), p. 102884. DOI: 10.1016/j.jnca.2020.102884.

[46] Pablo G. Madoery, Ramiro Detke, Lucas Blanco, Sandro Comerci, **Juan A. Fraire**, Aldana M. González-Montoro, Juan Carlos Bellassai, Grisel Maribel Britos, Silvia María Ojeda, and Jorge M. Finochietto. “Feature selection for proximity estimation in COVID-19 contact tracing apps based on Bluetooth Low Energy (BLE)”. In: *Elsevier Pervasive and Mobile Computing* 77 (2021), p. 101474. DOI: 10.1016/j.pmcj.2021.101474.

[47] Andrea Nardin, Fabio Dovis, and **Juan A. Fraire**. “Empowering the Tracking Performance of LEO-Based Positioning by Means of Meta-Signals”. In: *IEEE Journal of Radio Frequency Identification* 5.3 (2021), pp. 244–253. DOI: 10.1109/jrfid.2021.3077082.

[48] Raydel Ortigueira, **Juan A. Fraire**, Alex Becerra, Tomás Ferrer, and Sandra Céspedes. “RESS-IoT: A Scalable Energy-Efficient MAC Protocol for Direct-to-Satellite IoT”. In: *IEEE Access* 9 (2021), pp. 164440–164453. DOI: 10.1109/access.2021.3134246.

[49] Fernando D. Raverta, **Juan A. Fraire**, Pablo G. Madoery, Ramiro A. Demasi, Jorge M. Finochietto, and Pedro R. D’Argenio. “Routing in Delay-Tolerant Networks under uncertain contact plans”. In: *Elsevier Ad Hoc Networks* 123 (2021), p. 102663. DOI: 10.1016/j.adhoc.2021.102663.

[50] Germán J Soldano, **Juan A. Fraire**, Jorge M Finochietto, and Rodrigo Quiroga. “COVID-19 mitigation by digital contact tracing and contact prevention (app-based social exposure warnings)”. In: *Nature Scientific Reports* 11.1 (2021), p. 14421. DOI: 10.1038/s41598-021-93538-5.

[51] Olivier De Jonckère and **Juan A. Fraire**. “A shortest-path tree approach for routing in space networks”. In: *China Communications* 17.7 (2020), pp. 52–66. DOI: 10.23919/j.cc.2020.07.005.

[52] **Juan A. Fraire**, Gilles Nies, Carsten Gerstacker, Holger Hermanns, Kristian Bay, and Morten Bisgaard. “Battery-Aware Contact Plan Design for LEO Satellite Constellations: The Ulloriaq Case Study”. In: *IEEE Transactions on Green Communications and Networking* 4.1 (2020), pp. 236–245. DOI: 10.1109/tgcn.2019.2954166.

[53] Shulei Gong, Hong Shen, Kanglian Zhao, Ruhai Wang, Xinggan Zhang, Tomaso de Cola, and **Juan A. Fraire**. “Network Availability Maximization for Free-Space Optical Satellite Communications”. In: *IEEE Wireless Communications Letters* 9.3 (2020), pp. 411–415. DOI: 10.1109/lwc.2019.2958804.

[54] Gregory Stock, **Juan A. Fraire**, Tobias Mömke, Holger Hermanns, Fakhri Babayev, and Eduardo Cruz. “Managing Fleets of LEO Satellites: Nonlinear, Optimal, Efficient, Scalable, Usable, and Robust”. In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 39.11 (2020), pp. 3762–3773. DOI: 10.1109/tcad.2020.3012751.

[55] Zhibo Yan, **Juan A. Fraire**, Kanglian Zhao, Hongcheng Yan, Pablo G. Madoery, Wenfeng Li, and Hong Yang. “Distributed Contact Plan Design for GNSSs”. In: *IEEE Transactions on Aerospace and Electronic Systems* 56.1 (2020), pp. 660–672. DOI: 10.1109/taes.2019.2917492.

[56] **Juan A. Fraire**, Marius Feldmann, Felix Walter, Elena Fantino, and Scott C. Burleigh. “Networking in Interstellar Dimensions: Communicating With TRAPPIST-1”. In: *IEEE Transactions on Aerospace and Electronic Systems* 55.4 (2019), pp. 1656–1665. DOI: 10.1109/taes.2018.2874149.

[57] Juan Antonio Clemente, Guillaume Hubert, **Juan A. Fraire**, Francisco J Franco, Francesca Villa, Solenne Rey, Maud Baylac, Helmut Puchner, Hortensia Mecha, and Raoul Velazco. “SEU characterization of three successive generations of COTS SRAMs at ultralow bias voltage to 14.2-MeV neutrons”. In: *IEEE Transactions on Nuclear Science* 65.8 (2018), pp. 1858–1865. DOI: 10.1109/tns.2018.2800905.

[58] **Juan A. Fraire**, Pablo G. Madoery, Amir Charif, and Jorge M. Finochietto. “On route table computation strategies in Delay-Tolerant Satellite Networks”. In: *Elsevier Ad Hoc Networks* 80 (2018), pp. 31–40. DOI: 10.1016/j.adhoc.2018.07.002.

[59] Pablo G. Madoery, **Juan A. Fraire**, and Jorge M. Finochietto. “Congestion management techniques for disruption-tolerant satellite networks”. In: *Wiley International Journal of Satellite Communications and Networking* 36.2 (2018), pp. 165–178. DOI: 10.1002/sat.1210.

[60] Alexandre Coelho, Raphaël Laurent, Miguel Solinas, **Juan A. Fraire**, Emmanuel Mazer, Nacer-Eddine Zergainoh, Said Karaoui, and Raoul Velazco. “On the robustness of stochastic

bayesian machines". In: *IEEE Transactions on Nuclear Science* 64.8 (2017), pp. 2276–2283. DOI: 10.1109/tns.2017.2678204.

[61] **Juan A. Fraire**, Pablo G. Madoery, Scott C. Burleigh, Marius Feldmann, Jorge M. Finochietto, Amir Charif, Nacer-Eddine Zergainoh, and Raoul Velazco. "Assessing Contact Graph Routing Performance and Reliability in Distributed Satellite Constellations". In: *Journal of Computer Networks and Communications* 2017 (2017), 2830542:1–2830542:18. DOI: 10.1155/2017/2830542.

[62] **Juan A. Fraire**, Pablo G. Madoery, Jorge M. Finochietto, and Guillermo Leguizamón. "An evolutionary approach towards contact plan design for disruption-tolerant satellite networks". In: *Elsevier Applied Soft Computing* 52 (2017), pp. 446–456. DOI: 10.1016/j.asoc.2016.10.023.

[63] **Juan A. Fraire**, Pablo A. Ferreyra, and Raoul Velazco. "GLOrbit: A 3D Satellite Orbit Propagator for Network Topology Analysis". In: *Revista de la Facultad de Ciencias Exactas, Físicas y Naturales (UNC)* 3.2 (2016), pp. 29–35. URL: https://revistas.unc.edu.ar/index.php/FCEFyN/article/view/13881.

[64] **Juan A. Fraire**, Pablo G. Madoery, and Jorge M. Finochietto. "Traffic-aware contact plan design for disruption-tolerant space sensor networks". In: *Elsevier Ad Hoc Networks* 47 (2016), pp. 41–52. DOI: 10.1016/j.adhoc.2016.04.007.

[65] Pablo G Madoery, **Juan A. Fraire**, and Jorge M Finochietto. "Analysis of communication strategies for earth observation satellite constellations". In: *IEEE Latin America Transactions* 14.6 (2016), pp. 2777–2782. DOI: 10.1109/tla.2016.7555254.

[66] **Juan A. Fraire** and Jorge M. Finochietto. "Design challenges in contact plans for disruption-tolerant satellite networks". In: *IEEE Communications Magazine* 53.5 (2015), pp. 163–169. DOI: 10.1109/mcom.2015.7105656.

[67] **Juan A. Fraire** and Jorge M. Finochietto. "Routing-aware fair contact plan design for predictable delay tolerant networks". In: *Elsevier Ad Hoc Networks* 25 (2015), pp. 303–313. DOI: 10.1016/j.adhoc.2014.07.006.

[68] **Juan A. Fraire**, Pablo G Madoery, and Jorge M Finochietto. "On the design and analysis of fair contact plans in predictable delay-tolerant networks". In: *IEEE Sensors Journal* 14.11 (2014), pp. 3874–3882. DOI: 10.1109/jsen.2014.2348917.

[69] **Juan A. Fraire**, Alejandro Ferreyra, and Carlos Marques. "OpenCL overview, implementation, and performance comparison". In: *IEEE Latin America Transactions* 11.1 (2013), pp. 274–280. DOI: 10.1109/tla.2013.6502816.

Conference Papers

[1] Alexander Y. Choquenaira-Florez, **Juan A. Fraire**, and Hervé Rivano. "To Transmit or Not: Tiny On-Device Transformers for Uplink Prediction in Direct-to-Satellite IoT". In: *IEEE Latin-American Conference on Communications (LATINCOM)*. To appear. 2026.

[2] Benoit Coeugnet, André-Luc Beylot, Vincent Deslandes, and **Juan A. Fraire**. "Fast Analytical Tuning of Repetition-Based MAC for IoT in Sparse LEO Constellations". In: *IEEE International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM)*. Short paper, to appear. 2026.

[3] Ramiro Detke, Pablo G. Madoery, Renato Cherini, **Juan A. Fraire**, and Jorge M. Finochietto. "Security-Aware Route Selection in DTNs: Towards Optimal Key Provisioning". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), Space-Terrestrial Internetworking Workshop (STINT)*. To appear. 2026.

[4] Arnau Dolz-Puig, Adrian Perez-Portero, **Juan A. Fraire**, and Anna Calveras. "Online Learning for Predictive Satellite Telemetry Analysis: Towards a Digital Twin for Constellation Management". In: *Joint European Conference on Networks and Communications and 6G Summit (EuCNC/6G Summit)*. 2026, pp. 60–67. DOI: 10.1109/eucnc/6gsummit68295.2026.11577319.

[5] **Juan A. Fraire**. “On the Distribution of DTN Reachability Information: A Quantitative Push-vs-Pull Analysis”. In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), Space-Terrestrial Internetworking Workshop (STINT)*. To appear. 2026.

[6] **Juan A. Fraire**, Camilo Rojas Milla, Fabio Patrone, and Mario Marchese. “MeteorNet: Continuous-Time Emulation Platform for Edge Intelligence in LEO Constellations”. In: *IEEE International Mediterranean Conference on Communications and Networking (MeditCom)*. IEEE. 2026, pp. 1–6. DOI: 10.1109/meditcom67211.2026.11641012.

[7] Oriol Fusté, **Juan A. Fraire**, Anna Calveras Augé, and Joan A. Ruiz-de-Azua. “Deploying QUIC in Challenged Networks: Profiling and Volume-Aware Congestion Control”. In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), Space-Terrestrial Internetworking Workshop (STINT)*. To appear. 2026.

[8] Diego Maldonado, Megumi Kaneko, **Juan A. Fraire**, Alexandre Guitton, Oana Iova, and Hervé Rivano. “Joint LoRa and LR-FHSS Resource Allocation Optimization in Direct-to-Satellite IoT Networks”. In: *IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM)*. 2026, pp. 183–192. DOI: 10.1109/wowmom69805.2026.00032.

[9] Robin Ohs, Gregory F. Stock, Andreas Schmidt, **Juan A. Fraire**, Jörg Ott, and Holger Hermanns. “Dark Clouds Rising in Low-Earth Orbit: On Environmental Limits to Massive Orbital AI”. In: *ACM Workshop on LEO Networking and Communication (LEO-NET), co-located with ACM SIGCOMM*. 2026, pp. 1960–1965. DOI: 10.1145/3789240.3827598.

[10] Simon Rink, Olivier De Jonckère, **Juan A. Fraire**, and Holger Hermanns. “CGR*: A* Search for Contact Graph Routing”. In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), Space-Terrestrial Internetworking Workshop (STINT)*. To appear. 2026.

[11] Camilo Rojas Milla, Nour Badini, Fabio Patrone, **Juan A. Fraire**, and Mario Marchese. “Distributed Reinforcement Learning for Resource Management in Satellite Edge Computing Systems”. In: *IEEE Aerospace Conference*. IEEE. 2026, pp. 1–8. DOI: 10.1109/aero66936.2026.11519789.

[12] Ivan Sanchez-Cordero, Angel Macias, Miguel A. Garcia-Sevilla, Juan Luis Herrera, Sergio Laso, Cristina Vicente-Chicote, **Juan A. Fraire**, Oscar Ledesma, David Bernal, Julian Fernandez, Jaime Galan-Jimenez, and Javier Berrocal. “ESC: An Emulation Framework for the Spatial Computing Continuum”. In: *IFIP Networking Conference (IFIP Networking)*. 2026, pp. 1–6. DOI: 10.23919/ifipnetworking70592.2026.11579263.

[13] Arnau Singla, Anna Calveras, **Juan A. Fraire**, and Joan A. Ruiz-de-Azua. “Constellation Management for Inter-Satellite Link Scheduling in NB-IoT Non-Terrestrial Networks”. In: *Joint European Conference on Networks and Communications and 6G Summit (EuCNC/6G Summit)*. 2026, pp. 1300–1307. DOI: 10.1109/eucnc/6gsummit68295.2026.11577303.

[14] Arnau Singla, Roger Pueyo Centelles, Anna Calveras, **Juan A. Fraire**, and Joan A. Ruiz-de-Azua. “RF-Emulated TN-NTN Testbed for Service Continuity in Emergency Communications”. In: *Joint European Conference on Networks and Communications and 6G Summit (EuCNC/6G Summit)*. 2026, pp. 1308–1315. DOI: 10.1109/eucnc/6gsummit68295.2026.11577622.

[15] Shota Suzuki, **Juan A. Fraire**, and Keisuke Uehara. “IR-CGR: Time-Aware Inter-Region Contact Graph Routing for Scalable Space DTNs”. In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), Space-Terrestrial Internetworking Workshop (STINT)*. To appear. 2026.

[16] Alejandro M. Toledo, Jason Gerard, Sandra Céspedes, and **Juan A. Fraire**. “Lifespan-Aware Contact Plan Design for Interplanetary Free-Space Optical Networks”. In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE), Space-Terrestrial Internetworking Workshop (STINT)*. To appear. 2026.

[17] Wesley Woo, Tenzin Samten Ukyab, **Juan A. Fraire**, Scott Shenker, Sylvia Ratnasamy, and Shaddi Hasan. “Beyond the Cone Model: Quantifying the Impact of Beam-Level Constraints on LEO Satellite Networks”. In: *ACM Workshop on LEO Networking and Communication*

(LEO-NET), co-located with ACM SIGCOMM. 2026, pp. 1887–1893. DOI: 10.1145/3789240.3827588.

[18] Hannaneh Barahouei Pasandi, Mohammad Hosseini, Sina Darabi, Franck Rousseau, and **Juan A. Fraire**. "Adaptive Orbital Direct-to-Device: Rethinking Satellite IoT Architecture". In: *ACM Workshop on Mobility in the Evolving Internet Architecture (MobiArch)*. ACM, 2025, pp. 38–43. DOI: 10.1145/3737897.3767293.

[19] Nicolò Benso, Alessandro Compagnoni, Gregory F. Stock, **Juan A. Fraire**, Gabriel Maiolini Capez, Camilla Ottaviani, Daniel Gaetano Riviello, Giacomo Verardo, Mirca Gargiulo, Leonardo Ospizio, Carla-Fabiana Chiasserini, and Roberto Garello. "Distributed On-Demand Routing for VLEO Constellations with 3-Terminal Inter-Satellite Links". In: *International Conference on Software, Telecommunications and Computer Networks (SoftCOM)*. IEEE, 2025, pp. 1–7. DOI: 10.23919/softcom66362.2025.11197383. URL: https://ieeexplore.ieee.org/document/11197383.

[20] Alice Le Bihan, Felix Flentge, and **Juan A. Fraire**. "Custody Transfer and Compressed Status Reporting for Bundle Protocol Version 7". In: *International Conference on Space Operations (SpaceOps)*. 2025. DOI: 10.82217/spaceops2025_151.

[21] Arnau Dolz-Puig, Oriol Fusté, Adrian Perez-Portero, **Juan A. Fraire**, Joan A. Ruiz-de-Azua, and Anna Calveras. "Design of a Digital Twin for NB-IoT Satellite Constellations". In: *International Communications Satellite Systems Conference (ICSSC)*. IET. 2025, pp. 105–111. DOI: 10.1049/icp.2025.4045.

[22] Rachel Dudukovich and **Juan A. Fraire**. "Evolution of Interplanetary Internet: Technology Demonstrations for Space Network Operations". In: *International Conference on Space Operations (SpaceOps)*. 2025. URL: https://ntrs.nasa.gov/citations/20250002486.

[23] Alexander Ylnner Choquenaira Florez, Robin Ohs, **Juan A. Fraire**, and Hervé Rivano. "FLoRaSat 2: Simulating Cross-Linked Direct-to-Satellite IoT LEO Constellations". In: *Advanced Satellite Multimedia Systems Conference and Signal Processing for Space Communications Workshop (ASMS/SPSC)*. IEEE, 2025, pp. 1–8. DOI: 10.1109/asms/spsc64465.2025.10946025.

[24] **Juan A. Fraire**, Larissa Suzuki, Scott Burleigh, Leigh Torgerson, and Vinton Cerf. "Towards a Unified Routing Framework for the Interplanetary Internet of Things". In: *International Conference on Space Operations (SpaceOps)*. 2025. DOI: 10.82217/spaceops2025_411.

[25] Oriol Fusté, **Juan A. Fraire**, Anna Calveras, and Joan A. Ruiz-de-Azua. "Performance Comparison of DTN-Inspired Approaches for Store-and-Forward NTNs". In: *International Astronautical Congress (IAC), Space Communications and Navigation Symposium*. IAF, 2025, pp. 481–491. DOI: 10.52202/083082-0057.

[26] Jason Gerard, **Juan A. Fraire**, and Sandra Céspedes. "Modeling Pointing, Acquisition, and Tracking Delays in Free-Space Optical Satellite Networks". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2025, pp. 1–6. DOI: 10.1109/wisee57913.2025.11229866.

[27] Pablo Ilabaca, Diego Maldonado, **Juan A. Fraire**, Samuel Montejo Sánchez, and Sandra Céspedes. "Network Size Estimation in DtS-IoT: A LoRaWAN Approach for Satellite Constellations". In: *IEEE International Conference on Communications (ICC)*. IEEE, 2025, pp. 5506–5511. DOI: 10.1109/icc52391.2025.11162046.

[28] Olivier De Jonckère, Longrui Ma, and **Juan A. Fraire**. "A-SABR: The Adaptive Library for Schedule-Aware Bundle Routing". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2025, pp. 1–6. DOI: 10.1109/wisee57913.2025.11229852.

[29] Marcos Rojas Mardones, **Juan A. Fraire**, Oana Iova, Florent Dobler, Olivier Alphand, Didier Donsez, and Martin Heusse. "From the City to the Clouds: An Experimental Performance Evaluation of LR-FHSS". In: *International Conference on Distributed Computing in Smart*

Systems and the Internet of Things (DCOSS-IoT). 2025, pp. 246–250. DOI: 10.1109/dcoss-iot65416.2025.00041.

[30] Valentin Negrelli, Renato Cherini, and **Juan A. Fraire**. "Deep Reinforcement Learning for Routing in Uncertain DTNs with Graph Neural Networks". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2025, pp. 1–6. DOI: 10.1109/wisee57913.2025.11229839.

[31] Robin Ohs, Gregory F Stock, Andreas Schmidt, **Juan A. Fraire**, and Holger Hermanns. "Dirty Bits in Low-Earth Orbit: The Carbon Footprint of Launching Computers". In: *ACM SIGENERGY Energy Informatics Review* 5.2 (2025), pp. 26–33. DOI: 10.1145/3757892.3757896.

[32] Robin Ohs, Gregory F. Stock, **Juan A. Fraire**, Holger Hermanns, and Andreas Schmidt. "PhantomLink: Emulating Virtual End-to-End Links on Ground and in Orbit". In: *ACM/IRTF Applied Networking Research Workshop (ANRW)*. ACM, 2025, pp. 39–46. DOI: 10.1145/3744200.3744758.

[33] Camilla Ottaviani, Alessandro Compagnoni, **Juan A. Fraire**, Giacomo Verardo, Gabriel Maiolini Capez, Daniel Gaetano Riviello, Gregory Stock, Carla-Fabiana Chiasserini, and Roberto Garello. "Advanced Routing Strategies for LEO and VLEO Constellations: Ensuring Polar Coverage". In: *Advanced Satellite Multimedia Systems Conference and Signal Processing for Space Communications Workshop (ASMS/SPSC)*. IEEE, 2025, pp. 1–8. DOI: 10.1109/asms/spsc64465.2025.10946062.

[34] Hannaneh B Pasandi, Zhaowei Tan, Mohammad Hosseini, Sina Darabi, Franck Rousseau, and **Juan A. Fraire**. "FedSat-LAM: Enabling Large AI Models on Resource-Constrained Satellites via Hierarchical Federated Learning". In: *ACM International Workshop on Edge and Mobile Foundation Models (EdgeFM)*. ACM, 2025, pp. 19–24. DOI: 10.1145/3737902.3768355.

[35] Camilo Rojas, **Juan A. Fraire**, Fabio Patrone, and Mario Marchese. "On the Latency Trade-off Between Space and Terrestrial Clouds in Non-Terrestrial Networks". In: *Advanced Satellite Multimedia Systems Conference and Signal Processing for Space Communications Workshop (ASMS/SPSC)*. IEEE, 2025, pp. 1–8. DOI: 10.1109/asms/spsc64465.2025.10946050.

[36] Gregory F Stock, **Juan A. Fraire**, Holger Hermanns, Jedrzej Mosiezny, Yusra Al-Khazraji, Julio Ramirez Molina, and Evridiki V Ntagiou. "On the Role of AI in Managing Satellite Constellations: Insights from the ConstellAI Project". In: *International Conference on Space Operations (SpaceOps)*. 2025. DOI: 10.82217/spaceops2025_140.

[37] Gregory F. Stock, Alexander Haberl, **Juan A. Fraire**, and Holger Hermanns. "POMDP-Based Routing for DTNs with Partial Knowledge and Dependent Failures". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2025, pp. 1–6. DOI: 10.1109/wisee57913.2025.11229850.

[38] Wesley Woo, **Juan A. Fraire**, Sylvia Ratnasamy, Scott Shenker, and Shaddi Hasan. "Anyone, Anywhere, not Everyone, Everywhere: Starlink Doesn't End the Digital Divide". In: *ACM Workshop on Hot Topics in Networks (HotNets)*. Ed. by Alan Zaoxing Liu, Philip Brighten Godfrey, and Barath Raghavan. ACM, 2025, pp. 114–122. DOI: 10.1145/3772356.3772383.

[39] William Bernardoni, Robert Kassouf-Short, Robert Cardona, Brian Heller, Justin Curry, David Spivak, and **Juan A. Fraire**. "Network Storage Analysis via Semiring Geometry". In: *IEEE Aerospace Conference*. IEEE. 2024, pp. 1–19. DOI: 10.1109/aero58975.2024.10521207.

[40] Carlos E. Budde, Pedro R. D'Argenio, **Juan A. Fraire**, Arnd Hartmanns, and Zhen Zhang. "Modest Models and Tools for Real Stochastic Timed Systems". In: *Principles of Verification: Cycling the Probabilistic Landscape (Springer LNCS Festschrift)*. Ed. by Nils Jansen, Sebastian Junges, Benjamin Lucien Kaminski, Christoph Matheja, Thomas Noll, Tim Quatmann, Mariëlle Stoelinga, and Matthias Volk. Vol. 15261. Lecture Notes in Computer Science. Springer, 2024, pp. 115–142. DOI: 10.1007/978-3-031-75775-4_6.

[41] Ronny L. Bull, Rachel M. Dudukovich, **Juan A. Fraire**, Nadia Kortas, Robert Kassouf-Short, Aaron Smith, and Ethan Schweinsberg. "Network Emulation Testbed Capabilities for Pro-

totyping Space DTN Software and Protocols". In: *IEEE INFOCOM Workshops*. IEEE, 2024, pp. 1–8. DOI: 10.1109/infocomwkshps61880.2024.10620672.

[42] Benoit Coeugnet, Pablo G. Madoery, **Juan A. Fraire**, Jorge M. Finochietto, and Halim Yanikomeroglu. "Routing in All-Optical LEO Constellations with High-Altitude Ground Stations". In: *IEEE Middle East Conference on Communications and Networking (MECOM)*. IEEE, 2024, pp. 157–162. DOI: 10.1109/mecom61498.2024.10880869.

[43] Olivier De Jonckere, **Juan A. Fraire**, and Scott Burleigh. "Distributed Volume Management in Space DTNs: Scoping Schedule-Aware Bundle Routing". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2024, pp. 125–134. DOI: 10.1109/smc-it61443.2024.00021.

[44] Elena Fernández-Niño, **Juan A. Fraire**, Sergi Figuerola, Adriano Camps, Joan F. Munoz-Martin, and Joan A. Ruiz-de-Azua. "Design and Implementation of the Protocol Stack of the Control Plane in Hybrid Inter-Satellite Link Terminals". In: *International Astronautical Congress (IAC), Space Communications and Navigation Symposium*. IAF, 2024, pp. 665–670. DOI: 10.52202/078363-0071.

[45] Elena Fernández-Niño, **Juan A. Fraire**, Sergi Figuerola, Adriano Camps, and Joan Adrià Ruiz-de-Azua. "A Case Study of an Hybrid RF and Optical Inter-Satellite Link Terminal to Enhance Optical Pointing". In: *Joint European Conference on Networks and Communications and 6G Summit (EuCNC/6G Summit)*. IEEE, 2024, pp. 640–645. DOI: 10.1109/eucnc/6gsummit60053.2024.10597073.

[46] Jason Gerard, **Juan A. Fraire**, and Sandra Céspedes. "Unlocking DtS-IoT Medium Access Through the Massively Scalable Distributed Queuing Protocol". In: *IEEE World Forum on Internet of Things (WF-IoT)*. IEEE, 2024, pp. 840–845. DOI: 10.1109/wf-iot62078.2024.10811352.

[47] Jason Gerard, **Juan A. Fraire**, Raydel Ortigueira, and Sandra Céspedes. "DtS-IoT Resource Allocation Analysis Framework: Assessing DQ and RESS-IoT". In: *IEEE World Forum on Internet of Things (WF-IoT)*. IEEE, 2024, pp. 672–677. DOI: 10.1109/wf-iot62078.2024.10811213.

[48] Jason Gerard, Andre Ibrahim, **Juan A. Fraire**, and Sandra Céspedes. "Autonomous Max-Flow Interplanetary Laser Link Scheduling for Martian Exploration". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2024, pp. 70–75. DOI: 10.1109/wisee61249.2024.10850428.

[49] Alice Le Bihan, **Juan A. Fraire**, Pierre Francois, and Felix Flentge. "IPN-V: The Interplanetary Network Visualizer". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2024, pp. 142–150. DOI: 10.1109/smc-it61443.2024.00023.

[50] Pablo G. Madoery, **Juan A. Fraire**, Jorge M. Finochietto, Halim Yanikomeroglu, and Gunes Karabulut Kurt. "A Novel Non-Terrestrial Networks Architecture: All Optical LEO Constellations with High-Altitude Ground Stations". In: *IEEE International Conference on Communications Workshops (ICC Workshops)*. IEEE, 2024, pp. 1115–1120. DOI: 10.1109/iccworkshops59551.2024.10615868.

[51] Sebastián I Montoya, Diego Maldonado, **Juan A. Fraire**, and Sandra Céspedes. "On the Role of Delay Tolerant Networks and Contact Graph Routing in Direct-to-Satellite IoT". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2024, pp. 151–160. DOI: 10.1109/smc-it61443.2024.00024.

[52] Evridiki Ntagiou, Pierre Choukroun, **Juan A. Fraire**, Holger Hermanns, Petr Kubasta, Gregory Stock, Rafael José Badell Fabelo, Yusra Al-Khazraji, Sepideh Rahimian, Septika Pedi Artati, Jeremy Pierce-Mayer, Erica Rapp, Pierrick Houédé, Franck Appaix, and Marc Spigai. "Constellation Autonomy: AI Solutions for Adaptable and Efficient Operations". In: *International*

Astronautical Congress (IAC), Space Operations Symposium. IAF, 2024, pp. 303–316. DOI: 10.52202/078367-0034.

[53] Hannaneh B. Pasandi, **Juan A. Fraire**, Sylvia Ratnasamy, and Hervé Rivano. "A Survey on Direct-to-Device Satellite Communications: Advances, Challenges, and Prospects". In: *ACM Workshop on LEO Networking and Communication (LEO-NET)*. ACM, 2024, pp. 7–12. DOI: 10.1145/3697253.3697265.

[54] Camilo Rojas, **Juan A. Fraire**, Fabio Patrone, and Mario Marchese. "Advanced Constellation Emulation and Synthetic Datasets Generation for Non-Terrestrial Networks". In: *IEEE International Mediterranean Conference on Communications and Networking (MeditCom)*. IEEE, 2024, pp. 37–43. DOI: 10.1109/meditcom61057.2024.10621248.

[55] Camilo Rojas, **Juan A. Fraire**, Fabio Patrone, and Mario Marchese. "Fuzzy Logic-Based Orchestration of Multi-Access Edge Computing in LEO Satellite Constellations". In: *IEEE International Conference on Communications Workshops (ICC Workshops)*. IEEE, 2024, pp. 1685–1690. DOI: 10.1109/iccworkshops59551.2024.10615305.

[56] Germán L Scapin, Guido Alvarez, Nicolás Quiroga, Federico Collado, and **Juan A. Fraire**. "Performance Assessment of LoRaWAN DtS-IoT Networks: Operation Modes and Server Placement". In: *IEEE Biennial Congress of Argentina (ARGENCON)*. IEEE. 2024, pp. 1–8. DOI: 10.1109/argencon62399.2024.10735884.

[57] Arnau Singla, Anna Calveras, Francesc Betorz, Joan A. Ruiz-de-Azua, **Juan A. Fraire**, and Joan F. Munoz-Martin. "Integrating Satellite and Network Operations for NB-IoT NTN Connectivity: In-Orbit Test Design of the Constellation Management System". In: *International Astronautical Congress (IAC), Space Operations Symposium*. IAF, 2024, pp. 738–749. DOI: 10.52202/078367-0079.

[58] Arnau Singla, Franco Criscola, David Canales, Anna Calveras, **Juan A. Fraire**, and Joan A. Ruiz-de-Azua. "Integrating Satellite Constellation and Mobile Operations for Non-Terrestrial Networks: Preliminary Results of Dynamic Scheduling". In: *International Conference on Advances in Satellite and Space Communications (SPACOMM)*. 2024, pp. 8–13. URL: http://hdl.handle.net/2117/415648.

[59] Gregory Stock, **Juan A. Fraire**, Santiago Henn, Holger Hermanns, and Andreas Schmidt. "A Stability-First Approach to Running TCP Over Starlink". In: *IEEE International Conference on Communications Workshops (ICC Workshops)*. IEEE, 2024, pp. 1708–1713. DOI: 10.1109/iccworkshops59551.2024.10615714.

[60] Felix Walter, Marius Feldmann, **Juan A. Fraire**, and Scott Burleigh. "The Architectural Refinement of μD3TN: Toward a Software-Defined DTN Protocol Stack". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2024, pp. 161–170. DOI: 10.1109/smc-it61443.2024.00025.

[61] **Juan A. Fraire**, Alexandre Guitton, and Oana Iova. "LR-FHSS: j'ai perdu l'entête". In: *Rencontres Francophones sur la Conception de Protocoles (CoRes)*. 2023. URL: https://hal.science/hal-04087040v1.

[62] **Juan A. Fraire**, Alexandre Guitton, and Oana Iova. "Recovering Headerless Frames in LR-FHSS". In: *International Conference on Embedded Wireless Systems and Networks (EWSN)*. Ed. by Giancarlo Fortino, Valeria Loscrı`, Fabrizio Granelli, Tarek F. Abdelzaher, Antonella Guzzo, and Claudio Savaglio. ACM, 2023, pp. 170–180. URL: https://dl.acm.org/doi/10.5555/3639940.3639963.

[63] Olivier De Jonckère and **Juan A. Fraire**. "Inter-Regional Routing in Interplanetary Networks with Shortcuts and Contact Passageways". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2023, pp. 87–92. DOI: 10.1109/wisee58383.2023.10289358.

[64] Olivier De Jonckère, **Juan A. Fraire**, and Scott Burleigh. "Enhanced Pathfinding and Scalability with Shortest-Path Tree Routing for Space Networks". In: *IEEE International Conference*

on Communications (ICC). IEEE, 2023, pp. 4082–4088. DOI: 10.1109/icc45041.2023.10279514.

[65] Olivier De Jonckère, **Juan A. Fraire**, and Scott Burleigh. "On the Tractability of Yen's Algorithm and Contact Graph Modeling in Contact Graph Routing". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2023, pp. 80–86. DOI: 10.1109/wisee58383.2023.10289594.

[66] Diego Maldonado, **Juan A. Fraire**, Pablo Ilabaca, Hervé Rivano, and Sandra Céspedes. "Network Size Estimation for LoRa-Based Direct-to-Satellite IoT". In: *IEEE Cognitive Communications for Aerospace Applications Workshop (CCAAW)*. IEEE. 2023, pp. 1–6. DOI: 10.1109/ccaaw57883.2023.10219363.

[67] Matías Olmedo, **Juan A. Fraire**, Renato Cherini, Fabio Patrone, and Mario Marchese. "Routing in Scalable Delay-Tolerant Space Networks with Graph Neural Networks". In: *International Conference on Embedded Wireless Systems and Networks (EWSN)*. Ed. by Giancarlo Fortino, Valeria Loscrı`, Fabrizio Granelli, Tarek F. Abdelzaher, Antonella Guzzo, and Claudio Savaglio. ACM, 2023, pp. 211–218. URL: https://dl.acm.org/doi/10.5555/3639940.3639967.

[68] Camilo Rojas, **Juan A. Fraire**, Fabio Patrone, Alberto Gotta, and Mario Marchese. "Continuous Time Emulation for Software-Defined Non-Terrestrial Edge Computing Networks". In: *European Wireless Conference*. VDE. 2023, pp. 230–235. URL: https://ieeexplore.ieee.org/abstract/document/10461445.

[69] Pedro R. D'Argenio, **Juan A. Fraire**, Arnd Hartmanns, and Fernando D. Raverta. "Comparing Statistical and Analytical Routing Approaches for Delay-Tolerant Networks". In: *International Conference on Quantitative Evaluation of Systems (QEST)*. Ed. by Erika Ábrahám and Marco Paolieri. Vol. 13479. Lecture Notes in Computer Science. Springer, 2022, pp. 337–355. DOI: 10.1007/978-3-031-16336-4_17.

[70] **Juan A. Fraire**, Pablo G. Madoery, Mehdi Ait Mesbah, Oana Iova, and Fabrice Valois. "Simulating LoRa-Based Direct-to-Satellite IoT Networks with FLoRaSaT". In: *IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM)*. IEEE, 2022, pp. 464–470. DOI: 10.1109/wowmom54355.2022.00072.

[71] **Juan A. Fraire**, Mehdi Mesbah, Pablo Madoery, Oana Iova, and Fabrice Valois. "FLoRaSat: Le simulateur pour la communication LoRa avec les satellites LEO". In: *Rencontres Francophones sur la Conception de Protocoles (CoRes)*. 2022, pp. 1–4. URL: https://hal.science/hal-03657890v1.

[72] Gregory Stock, **Juan A. Fraire**, and Holger Hermanns. "Distributed On-Demand Routing for LEO Mega-Constellations: A Starlink Case Study". In: *Advanced Satellite Multimedia Systems Conference and Signal Processing for Space Communications Workshop (ASMS/SPSC)*. IEEE, 2022, pp. 1–8. DOI: 10.1109/asms/spsc55670.2022.9914716.

[73] Guido Alvarez, Mayco Dagatti, Matías Olmedo, Neyra Poveda, and **Juan A. Fraire**. "Satellite-Based AIS Trade-off Analysis in the Context of the PANSAT CubeSat Mission". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2021, pp. 77–84. DOI: 10.1109/smc-it51442.2021.00017.

[74] Gregory Stock, **Juan A. Fraire**, Holger Hermanns, Eduardo Cruz, Alastair Isaacs, and Zhana Imbrosh. "On the automation, optimization, and in-orbit validation of intelligent satellite constellation operations". In: *AIAA/USU Conference on Small Satellites (SmallSat)*. 2021. URL: https://digitalcommons.usu.edu/smallsat/2021/all2021/168/.

[75] Kai Vogelgesang, **Juan A. Fraire**, and Holger Hermanns. "Uplink Transmission Probability Functions for LoRa-Based Direct-to-Satellite IoT: A Case Study". In: *IEEE Global Communications Conference (GLOBECOM)*. IEEE, 2021, pp. 1–6. DOI: 10.1109/globecom46510.2021.9685152.

[76] Pedro R. D'Argenio, **Juan A. Fraire**, and Arnd Hartmanns. "Sampling Distributed Schedulers for Resilient Space Communication". In: *NASA Formal Methods Symposium (NFM)*. Ed. by

Ritchie Lee, Susmit Jha, and Anastasia Mavridou. Vol. 12229. Lecture Notes in Computer Science. Springer, 2020, pp. 291–310. DOI: 10.1007/978-3-030-55754-6_17.

[77] Alexis Alfredo Dowhuszko, **Juan A. Fraire**, Musbah Shaat, and Ana I. Pérez-Neira. "LEO satellite constellations to offload optical terrestrial networks in placement of popular content in 5G edge nodes". In: *International Conference on Transparent Optical Networks (ICTON)*. IEEE, 2020, pp. 1–6. DOI: 10.1109/icton51198.2020.9203447.

[78] **Juan A. Fraire**, Santiago Henn, Fabio Dovis, Roberto Garello, and Giorgio Taricco. "Sparse Satellite Constellation Design for LoRa-based Direct-to-Satellite Internet of Things". In: *IEEE Global Communications Conference (GLOBECOM)*. IEEE, 2020, pp. 1–6. DOI: 10.1109/globecom42002.2020.9348042.

[79] Andrea Nardin, Fabio Dovis, and **Juan A. Fraire**. "Empowering the Tracking Performance of LEO PNT by Means of Meta-Signals". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2020, pp. 153–158. DOI: 10.1109/wisee44079.2020.9262698.

[80] Blas F Vega and **Juan A. Fraire**. "Experimental Evaluation of On-Board Contact-Graph Routing Solutions for Future Nano-Satellite Constellations". In: *AIAA/USU Conference on Small Satellites (SmallSat)*. 2020. URL: https://digitalcommons.usu.edu/smallsat/2020/all2020/56/.

[81] Pablo A. Ferreyra, **Juan A. Fraire**, Fabian Gomez, Raoul Velazco, Daniel Sánchez, and Dardo Vinas Viscardi. "Delay-Tolerant Wireless Networks on Chip: Preliminary Analysis and Results". In: *IEEE Latin American Test Symposium (LATS)*. IEEE, 2019, pp. 1–6. DOI: 10.1109/latw.2019.8704623.

[82] **Juan A. Fraire**, Sandra Céspedes, and Nicola Accettura. "Direct-To-Satellite IoT - A Survey of the State of the Art and Future Research Perspectives - Backhauling the IoT Through LEO Satellites". In: *International Conference on Ad-Hoc Networks and Wireless (ADHOC-NOW)*. Ed. by Maria Rita Palattella, Stefano Scanzio, and Sinem Coleri Ergen. Vol. 11803. Lecture Notes in Computer Science. Springer, 2019, pp. 241–258. DOI: 10.1007/978-3-030-31831-4_17.

[83] Santiago M Henn, **Juan A. Fraire**, Pablo C Marino, Maximiliano Fischer, Andrea Bianco, and Holger Hermanns. "Towards the design of space missions with optimal accessibility via relay satellites". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2019, pp. 100–105. DOI: 10.1109/smc-it.2019.00018.

[84] Lu Liu, Ruhai Wang, Yan Zhou, Wenfeng Li, Yi Jin, and **Juan A. Fraire**. "Energy consumption of bundle protocol for reliable file delivery over deep-space channels". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2019, pp. 106–111. DOI: 10.1109/smc-it.2019.00019.

[85] Daniel Sánchez, Pablo A. Ferreyra, **Juan A. Fraire**, Fabian Gomez, Raoul Velazco, and Dardo Vinas Viscardi. "A Wireless Embedded System for Measuring the Effects of Ionizing Radiations". In: *IEEE Latin American Test Symposium (LATS)*. IEEE, 2019, pp. 1–6. DOI: 10.1109/latw.2019.8704607.

[86] Yan Zhou, Ruhai Wang, Lu Liu, Kanglian Zhao, Wenfeng Li, Xiangyu Lin, and **Juan A. Fraire**. "Effect of Link Disruption on BP/LTP over a Relay-based Communication Architecture for International Space Station". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2019, pp. 95–99. DOI: 10.1109/smc-it.2019.00017.

[87] Bin Cao, Ruhai Wang, Alaa Sabbagh, Siwei Peng, Kanglian Zhao, **Juan A. Fraire**, Guannan Yang, and Yue Wang. "Expected File-Delivery Time of DTN Protocol over Asymmetric Space Internetwork Channels". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2018, pp. 147–151. DOI: 10.1109/wisee.2018.8637352.

[88] Michael Carosino, **Juan A. Fraire**, and James A. Ritcey. "Integrating Scheduled DTNs and TDMA-based MAC Sublayers: Preliminary Results". In: *IEEE International Conference on*

Wireless for Space and Extreme Environments (WiSEE). IEEE, 2018, pp. 141–146. DOI: 10.1109/wisee.2018.8637344.

[89] Alexandre Coelho, Amir Charif, Nacer-Eddine Zergainoh, **Juan A. Fraire**, and Raoul Velazco. "A soft-error resilient route computation unit for 3D Networks-on-Chips". In: *Design, Automation and Test in Europe Conference (DATE)*. Ed. by Jan Madsen and Ayse K. Coskun. IEEE, 2018, pp. 1357–1362. DOI: 10.23919/date.2018.8342224.

[90] Marius Feldmann, **Juan A. Fraire**, and Felix Walter. "Tracking Lunar Ring Road Communication". In: *IEEE International Conference on Communications (ICC)*. IEEE, 2018, pp. 1–7. DOI: 10.1109/icc.2018.8423031.

[91] **Juan A. Fraire**, Gilles Nies, Holger Hermanns, Kristian Bay, and Morten Bisgaard. "Battery-Aware Contact Plan Design for LEO Satellite Constellations: The Ulloriaq Case Study". In: *IEEE Global Communications Conference (GLOBECOM)*. IEEE, 2018, pp. 1–7. DOI: 10.1109/glocom.2018.8647822.

[92] Pablo G. Madoery, **Juan A. Fraire**, Fernando D. Raverta, and Jorge M. Finochietto. "On the Trade-off Analysis of Centralized and Distributed Routing in Delay Tolerant Satellite Networks". In: *School on Systems and Networks (SSN)*. Ed. by Sandra Céspedes and Javier Bustos-Jiménez. Vol. 2178. CEUR Workshop Proceedings. CEUR-WS.org, 2018, pp. 89–93. URL: https://ceur-ws.org/Vol-2178/SSN2018%5C_paper%5C_32.pdf.

[93] Pablo G. Madoery, **Juan A. Fraire**, Fernando D. Raverta, Jorge M. Finochietto, and Scott C. Burleigh. "Managing Routing Scalability in Space DTNs". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2018, pp. 177–182. DOI: 10.1109/wisee.2018.8637324.

[94] Pablo G. Madoery, Fernando D. Raverta, **Juan A. Fraire**, and Jorge M. Finochietto. "Routing in Space Delay Tolerant Networks under Uncertain Contact Plans". In: *IEEE International Conference on Communications (ICC)*. IEEE, 2018, pp. 1–6. DOI: 10.1109/icc.2018.8422917.

[95] Fernando D. Raverta, Ramiro Demasi, Pablo G. Madoery, **Juan A. Fraire**, Jorge M. Finochietto, and Pedro R. D'Argenio. "A Markov Decision Process for Routing in Space DTNs with Uncertain Contact Plans". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2018, pp. 189–194. DOI: 10.1109/wisee.2018.8637330.

[96] Fernando D. Raverta, Pablo G. Madoery, **Juan A. Fraire**, and Jorge M. Finochietto. "Modeling Routing in Scheduled Delay Tolerant Networks Under Uncertainties". In: *School on Systems and Networks (SSN)*. Ed. by Sandra Céspedes and Javier Bustos-Jiménez. Vol. 2178. CEUR Workshop Proceedings. CEUR-WS.org, 2018, pp. 94–97. URL: https://ceur-ws.org/Vol-2178/SSN2018%5C_paper%5C_33.pdf.

[97] Yue Wang, Kanglian Zhao, Wenfeng Li, **Juan A. Fraire**, Zhili Sun, and Yuan Fang. "Performance Evaluation of QUIC with BBR in Satellite Internet". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2018, pp. 195–199. DOI: 10.1109/wisee.2018.8637347.

[98] JA Clemente, **Juan A. Fraire**, M Solinas, FJ Franco, Francesca Villa, Solenne Rey, Maud Baylac, Helmut Puchner, H Mecha, and Raoul Velazco. "SEU sensitivity trends of three successive generations of COTS SRAMs at ultra low bias voltage". In: *Conference on Radiation Effects on Components and Systems (RADECS)*. 2017. URL: https://in2p3.hal.science/in2p3-01735635v1.

[99] **Juan A. Fraire**. "Introducing contact plan designer: A planning tool for DTN-based space-terrestrial networks". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2017, pp. 124–127. DOI: 10.1109/smc-it.2017.28.

[100] **Juan A. Fraire**, Marius Feldmann, and Scott C. Burleigh. "Benefits and challenges of cross-linked ring road satellite networks: A case study". In: *IEEE International Conference on Communications (ICC)*. IEEE, 2017, pp. 1–7. DOI: 10.1109/icc.2017.7996778.

[101] **Juan A. Fraire**, Pablo Madoery, Fernando Raverta, Jorge M Finochietto, and Raoul Velazco. "Dtnsim: Bridging the gap between simulation and implementation of space-terrestrial dtns". In: *IEEE International Conference on Space Mission Challenges for Information Technology (SMC-IT)*. IEEE. 2017, pp. 120–123. DOI: 10.1109/smc-it.2017.27.

[102] Pablo G. Madoery, Fernando D. Raverta, **Juan A. Fraire**, and Jorge M. Finochietto. "On the Performance Analysis of Disruption Tolerant Satellite Networks under Uncertainties". In: *Workshop on Information Processing and Control (RPIC)*. IEEE, 2017, pp. 1–6. DOI: 10.23919/rpic.2017.8214346.

[103] M. Solinas, Alexandre Coelho, **Juan A. Fraire**, Nacer-Eddine Zergainoh, Pablo A. Ferreyra, and Raoul Velazco. "Preliminary results of NETFI-2: An automatic method for fault injection on HDL-based designs". In: *IEEE Latin American Test Symposium (LATS)*. IEEE, 2017, pp. 1–4. DOI: 10.1109/latw.2017.7906748.

[104] A. Coelho, M. Solinas, R. Laurent, **Juan A. Fraire**, E. Mazer, Nacer-Eddine Zergainoh, S. Karaoui, and Raoul Velazco. "Evidences of Stochastic Bayesian Machines Robustness Against SEUs and SETs". In: *Conference on Radiation Effects on Components and Systems (RADECS)*. 2016, pp. 1–5. DOI: 10.1109/radecs.2016.8093156.

[105] **Juan A. Fraire**, Pablo G. Madoery, Jorge M. Finochietto, Pablo A. Ferreyra, and Raoul Velazco. "Internetworking approaches towards along-track segmented satellite architectures". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2016, pp. 123–128. DOI: 10.1109/wisee.2016.7877316.

[106] **Juan A. Fraire** and Esteban G Kocian. "A Flexible and Autonomous Communication Subsystem for Constellations of Small-Satellites". In: *AIAA/USU Conference on Small Satellites (SmallSat)*. 2015. URL: https://digitalcommons.usu.edu/smallsat/2015/all2015/86/.

[107] **Juan A. Fraire**, Pablo G. Madoery, Jorge M. Finochietto, and Edward J. Birrane. "Congestion modeling and management techniques for predictable disruption tolerant networks". In: *IEEE Conference on Local Computer Networks (LCN)*. Ed. by Salil S. Kanhere, Jens Tölle, and Soumaya Cherkaoui. IEEE Computer Society, 2015, pp. 544–551. DOI: 10.1109/lcn.2015.7366369.

[108] **Juan A. Fraire**, Pablo G. Madoery, Jorge M. Finochietto, and Guillermo Leguizamón. "Preliminary results of an evolutionary approach towards Contact Plan design for satellite DTNs". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. Ed. by Ron Brown and Ali Abedi. IEEE, 2015, pp. 1–7. DOI: 10.1109/wisee.2015.7393097.

[109] Marcos Alfonzo, **Juan A. Fraire**, Esteban Kocian, and Nicolás Alvarez. "Development of a DTN bundle protocol convergence layer for SpaceWire". In: *IEEE Biennial Congress of Argentina (ARGENCON)*. IEEE. 2014, pp. 770–775. DOI: 10.1109/argencon.2014.6868586.

[110] **Juan A. Fraire** and Pablo A. Ferreyra. "Assessing DTN Architecture Reliability for Distributed Satellite Constellations: Preliminary Results from a Case Study". In: *IEEE Biennial Congress of Argentina (ARGENCON)*. IEEE, 2014, pp. 564–569. DOI: 10.1109/argencon.2014.6868551.

[111] **Juan A. Fraire**, Pablo G. Madoery, and Jorge M. Finochietto. "Leveraging routing performance and congestion avoidance in predictable delay tolerant networks". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2014, pp. 1–7. DOI: 10.1109/wisee.2014.6973079.

[112] **Juan A. Fraire**, Pablo Ferreyra, and Carlos Marques. "Opencl-accelerated simplified general perturbations 4 algorithm". In: *Argentine Symposium on Technology (AST), JAIIO*. 2013. URL: http://sedici.unlp.edu.ar/handle/10915/76850.

[113] **Juan A. Fraire**, Pablo G. Madoery, and Jorge M. Finochietto. "On the design of fair contact plans in predictable Delay-Tolerant Networks". In: *IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2013, pp. 1–7. DOI: 10.1109/wisee.2013.6737548.

Preprints

[1] Alexander Y. Choquenaira-Florez, Hervé Rivano, and **Juan A. Fraire**. "Transmit Less, Deliver More: Frugal On-Device Intelligence for Energy- and Spectrum-Sustainable Direct-to-Satellite IoT". In: *HAL preprint* (2026). NTN Innovation Competition 2026. URL: https://hal.science/hal-05800184.

[2] Huan Yan, **Juan A. Fraire**, Ziqi Yang, Kanglian Zhao, Wenfeng Li, Jinjun Zheng, Ziyu Wang, and Changgang Zheng. "Progressive Replacement Sequence Planning for Laser-Enhanced BeiDou Navigation Constellations". In: *arXiv preprint* (2026). DOI: 10.48550/arxiv.2607.25235. arXiv: 2607.25235.

[3] Marius Feldmann, Tobias Nöthlich, Felix Walter, Maximilian Nitsch, **Juan A. Fraire**, Georg A. Murzik, and Fiona Fuchs. "Selected Results from the REDMARS2 Project: Recursive Delay-Tolerant Networking using Bundle-in-Bundle Encapsulation". In: *arXiv preprint* (2025). DOI: 10.48550/arxiv.2510.27325. arXiv: 2510.27325.

[4] Santiago M. Henn, **Juan A. Fraire**, Nicola Accettura, Sandra Céspedes, and Holger Hermanns. "Multi-Gateway LoRaWAN Throughput Modeling in Direct-to-Satellite IoT Constellations". In: *HAL preprint* (2024). URL: https://hal.science/hal-04749023.

Research Supervision

- **Doctoral Students — Completed** (7):
  - **Jason Gerard** (2023–2026), Concordia University: co-supervised with Dr. Sandra Céspedes. Defended 29 June 2026 with distinction.
  - **Santiago Henn** (2022–2026), UNC / CONICET: sole supervision. Defended 19 June 2026; CONICET doctoral scholarship secured under my direction.
  - **Gabriel Maiolini Capez** (2021–2026), Politecnico di Torino: co-supervised with Prof. R. Garello. Defended 23 February 2026.
  - **Camilo Rojas Milla** (2022–2026), University of Genoa: co-supervised with Dr. Fabio Patrone. Defended 23 February 2026.
  - **Diego Maldonado** (2022–2026), INSA Lyon / Inria: co-supervised with Prof. H. Rivano. Defended 25 February 2026.
  - **Fernando Raverta** (2020–2024), UNC: co-supervised with Dr. J. Finochietto.
  - **Raydel Ortigueira** (2020–2023), Universidad de Chile: co-supervised with Dr. Sandra Céspedes.
- **Doctoral Students — Ongoing** (9):
  - **Alexander Ylnner Choquenaira Florez** (2024–2027), Inria: sole supervision.
  - **Gregory Stock** (2021–2026), Saarland University: co-supervised with Prof. Holger Hermanns.
  - **Elena F. Niño** (2023–2026), i2CAT / UPC: co-supervised with Dr. J. Ruiz-de-Azua.
  - **Arnau Singla** (2023–2026), i2CAT / UPC: co-supervised with Dr. J. Ruiz-de-Azua.
  - **Robin Ohs** (2023–2027), Saarland University: co-supervised with Prof. Holger Hermanns — lifecycle sustainability of in-orbit computing.
  - **Wenxiao Ge** (2024–2027), Nanjing University: co-supervised with Prof. Kanglian Zhao — Space Cloud architecture and design.
  - **Benoit Coeugnet** (2025–2028), IRIT / Inria: co-supervised with Dr. André-Luc Beylot.
  - **Alice Le Bihan** (2026–2029), INSA Lyon / Inria: co-supervised with Dr. Pierre François (INSA doctoral contract, École Doctorale InfoMaths, co-funded by ESA OSIP).
  - **Oriol Fusté** (2024–2027), i2CAT / UPC: co-supervised with Dr. J. Ruiz-de-Azua — DTN-inspired transport and QUIC for challenged space networks.
- **Postdoctoral**:
  - **Hannah B. Pasandi** (2024–2025), Inria: co-supervised with Prof. H. Rivano (INSA, Inria) and Prof. Sylvia Ratnasamy (UC Berkeley); now at UC Berkeley.
- **Master's & Engineering Theses** (25+):
  - Inria / INSA Lyon: Mehdi Ait Mesbah (2021), Sebastian Montoya Tapia (2022), Benoit Coeugnet

(2023–2024), Longrui Ma (2025, with Dr. O. De Jonckère), Lucas Poulhe (2025), Alice Le Bihan (2026), Pablo Illabaca, Diego Maldonado.
  - Inria Chile: Antonio Dávila (PUC Chile, 2026, with Dr. O. Iova).
  - FIUBA–CONAE (Master in Space Technology): Federico Gesualdo and Andrés Pozzi (since 2026, co-directed with Dr. S. Henn).
  - UNC (FAMAF / FCEFyN): Juan Costamagna (2021), Benjamin Martínez Picech (2023–2024), Valentín Negrelli (2024), Elías Gasparini, Dardo Ariel Viñas, Octavio Percivaldi, Nicole Wenzinger, Ulises Torrella, Roderick González, Denis Martínez, Ignacio Pintos Paladea, Fernando Blas, Pablo Soligo, Guido Ale.
  - Saarland University: Carsten Gerstacker (2019–2020), Kai Vogelgesang (2021), Robin Ohs, Simon Rink.
  - CONAE (Instituto Gulich): Blas Vega (2022–2023), Santiago Henn, Matías Olmedo, Marcos Alfonzo.
  - Politecnico di Torino: Camilla Ottaviani (2025–2026).
- **Undergraduate Final Projects**: Gonzalo Gigena and Guido Alvarez (UNC, 2022, satellite pose estimation), Alejandro Toledo (UNC, 2025, lifespan-aware link scheduling), Ignacio B. Ceballos (UNC, 2026, Bundle Protocol sockets), Tomás Maraschio (UNC, 2026, A* search in A-SABR), Esteban D. Druetta (UNC, 2026, dynamic fault trees for orbital data centres), Juan Sellanes Goncalves, Carlos Quiroga Ruiz and Leonel Garro Linck (UBA, 2024, lunar navigation).

Teaching

- **Lecturer, M.Sc. in Telecommunications**, INSA Lyon, France (Since 2021)
  - Course: *Satellite Communications and Navigation* (with externals from CNES).
- **Guest Professor, Computer Science**, Dependable Systems and Software, Saarland University, Germany (Since September 2018)
  - Courses: *Space Informatics* (block course), *Space Networking* (seminar).
- **Associate Professor, Computer Science**, FAMAF, Universidad Nacional de Córdoba (UNC), Argentina (Since May 2012)
  - *Computer Architecture* (May 2012 – December 2013); *Computer Networks* (since January 2014); *Operating Systems* (since August 2025).
- **Academic Advisor**, UFS-Gullich-CONAE:
  - Coordinator, Aerospace Industry and Systems specialization, UBA (Since January 2023): lead lectures on *Introduction to Space Systems* and *Space Systems Engineering*.
  - Director, Satellite Technology Master (MTS) (January 2020 – December 2022): led lectures on satellite communications and networking.

Invited Talks & Keynotes

- **Keynote, AIxNET 2026 (1st International Conference on AI and Interconnected Networks), Inria Paris, France** (November 2026, upcoming): "AI for Space, Space for AI: The Rise and Limits of the Space Cloud."
- **Inria Scientific Review (Revue Scientifique), NTN session, Paris, France** (November 2026, upcoming): "AI for Space, Space for AI: The Rise and Limits of the Space Cloud."
- **Cnam (Conservatoire National des Arts et Métiers), Paris, France** (October 2026): "AI for Space, Space for AI: The Rise and Limits of the Space Cloud."
- **MISSION Workshop at ETAPS 2026, Turin, Italy** (April 2026): "The Evolution of Uncertainty-Aware DTN Routing: MISSION-Driven Advances."
- **Dagstuhl Seminar "Connected Space: Challenges and Opportunities in Satellite Computing and Networking", Schloss Dagstuhl, Germany** (February 2026): "Research Opportunities and Challenges in Space and Satellite Networking."
- **FACAS 2026 (Taller Argentino de Fundamentos para el Análisis y la Construcción Automática**

de Software), Argentina (March 2026): "Demystifying Satellite Constellations: A Practical Guide for Computer Scientists."
- **Shenzhen Technology University (SZTU), Shenzhen, China** (March 2026): "Dependent Space: The Why, the How, and the Now for Space Systems Research."
- **Nanjing University, Nanjing, China** (March 2026): "Research Opportunities and Challenges in Space and Satellite Networking."
- **University of Sydney, Australia** (March 2026): "Research Opportunities and Challenges in Space and Satellite Networking."
- **Inria Chile, Santiago, Chile** (November 2025): "Research Opportunities and Challenges in Space and Satellite Networking."
- **RPIC 2025, San Francisco, Córdoba, Argentina** (October 2025): "Research Opportunities and Challenges in Space and Satellite Networking."
- **IEEE ICC Workshop on Satellite Mega-Constellations in the 6G Era, Denver, CO, USA** (June 2024): "On the Role of AI in Satellite Mega-Constellations."
- **UC Berkeley, NetSys Lab, Berkeley, CA, USA** (2024): "Interplanetary Internet of Things: Protocols and Simulation."
- **IEEE World Forum on Internet of Things (WF-IoT), Aveiro, Portugal** (October 2023): "Routing in the Interplanetary Internet of Things."
- **IEEE Cognitive Communications for Aerospace Applications (CCAAW), Cleveland, OH, USA** (July 2023): "Delay-Tolerant Networking Modeling and Experimentation."
- **i2Cat Workshop, Universidad Politécnica de Catalunya, Spain** (October 2023): "Interplanetary Internet of Things: Protocols, Routing and Simulation."
- **IEEE ARGENCON, Chaco, Argentina** (December 2020): "Towards a Space Internet: Experiences with the GomX-4 Mission."
- **International School on Radiation Effects for Embedded Systems (SERESSA), Virtual** (December 2020): "Transient Faults in Space Delay Tolerant Networking."
- **Nanjing University & Space Intelligent Networking Forum, Nanjing, China** (June 2018): "Motivation and Challenges in Space-Terrestrial Integrated Networks."
- **School of Systems and Networks, Valdivia, Chile** (October 2018): "Motivaciones y desafíos en redes integradas espacio-tierra."
- **International Conference of Satellite Technology (ICST), Bandung, Indonesia** (November 2017): "Routing, Congestion, and Planning in DTNs: Solutions and Challenges."
- **CCSDS in China Symposium, Nanjing, China** (June 2017): "Routing in Space Networks: Solutions and Challenges."
- **Space Symposium, Colorado Springs, USA** (April 2016): "Argentina in Space."

Professional Service & Leadership

Standards & Community Leadership

- **IETF/IRTF**: Chair, Systems and Protocol Aspects for Circumstellar Environments Research Group (SPACE RG, IRTF), member of the Internet Research Steering Group (IRSG); sessions chaired at IETF 124–126 and an interim meeting at ESA/ESOC (2026); author of Internet-Drafts on satellite-constellation registries and space-networking tools; active participant of the IETF DTN working group.
- **IPNSIG** (Interplanetary Networking Special Interest Group): Board Member (since 2024).
- **Internet Governance Forum (IGF)**: Academic Advisor to the Dynamic Coalition on the Interplanetary Internet.
- **CCSDS** (Consultative Committee for Space Data Systems): active participant of the SIS area.

Workshop & Conference Organization

- **Co-Founder & Chair, STINT (Space-Terrestrial Internetworking Workshop)** — 13 editions (2014–2026), co-located with IEEE WiSEE / SMC-IT:
  - Leuven 2026, Halifax 2025, Mountain View 2024, Aveiro 2023, Winnipeg 2022, California 2021,

Vicenza 2020, Pasadena 2019, Huntsville 2018, Alcalá de H. 2017, Aachen 2016, Orlando 2015, Laurel 2014.

- **IEEE World Forum on Internet of Things (WF-IoT) 2027**: Workshops Chair.
- **NTN Days 2026, Cnam, Paris, France**: Co-organizer (with V. Conan and P. Gu), October 2026.
- **MISSION Workshop at ETAPS 2026, Turin, Italy**: Co-organizer and program committee, April 2026.
- **SERESSA** (School on Radiation Effects on Embedded Systems for Space): Organizing committee, Córdoba 2024; Publicity chair and webmaster, Bariloche 2014.
- **ASMS/SPSC 2024, Sitges, Spain**: Chair, Space-Terrestrial Integrated IoT track (speakers from Sateliot, Kineis, Fossa).
- **IEEE ARGENCON 2020, Chaco, Argentina**: Chair, Aeronautics and Aerospace track.

Technical Program Committees & Reviewing

- **TPC Member**: IEEE Globecom, IEEE ICC (incl. 2026 workshops on Massive IoT and 6GSatComNet; ICC 2027 Satellite & Space Communications symposium), IEEE WiSEE, IEEE FNWF, SoftCOM, ACM CHANTS, IFIP Networking, IEEE LCN, WONS, AlgoTel/CoRes, CASE, MISSION@ETAPS, NTN'26 International Research Competition.
- **Journal Reviewer**: IEEE/ACM Trans. Networking, IEEE Trans. Aerospace and Electronic Systems, IEEE Trans. Vehicular Technology, IEEE Trans. Reliability, Elsevier Computer Networks, Ad-Hoc Networks and Array, Acta Astronautica, IEEE Communications Magazine/Letters, IEEE Networking Letters, IEEE Sensors Journal, IEEE Access, MDPI Sensors, JCOMSS, Springer Wireless Communications.
- **Ph.D. Juries**: T. Algarra Ulierte (TU Hamburg, reviewer, 2026); G. Maiolini Capez (Politecnico di Torino, final evaluation board, 2026); F. Dobler (Université Grenoble Alpes, examiner, 2026); doctoral progress committee at UTEM (Chile, 2026).
- **Other Evaluation**: IRTF travel-grant committee (IETF 127, 2026); research-grant reviewer (e.g., Argentina's PICT).

Outreach & Media

- **El Mercurio** (Chile): interview on security vulnerabilities of geostationary satellite links (2026).
- **FLISoL Córdoba**: workshop "Satélites para todos: diseñá constelaciones y redes espaciales con software libre" (April 2026, with S. Henn).
- **IPNSIG Promotional Video**: watch (July 2023).
- **El Gato y la Caja**: "Con el Ojo en el Cielo" (March 2019); "Rastis Satelitales" (October 2018); "Dame una Señal" (April 2018).
- **SpaceNews**: "Networking the Space-Terrestrial Frontier" (June 2018).